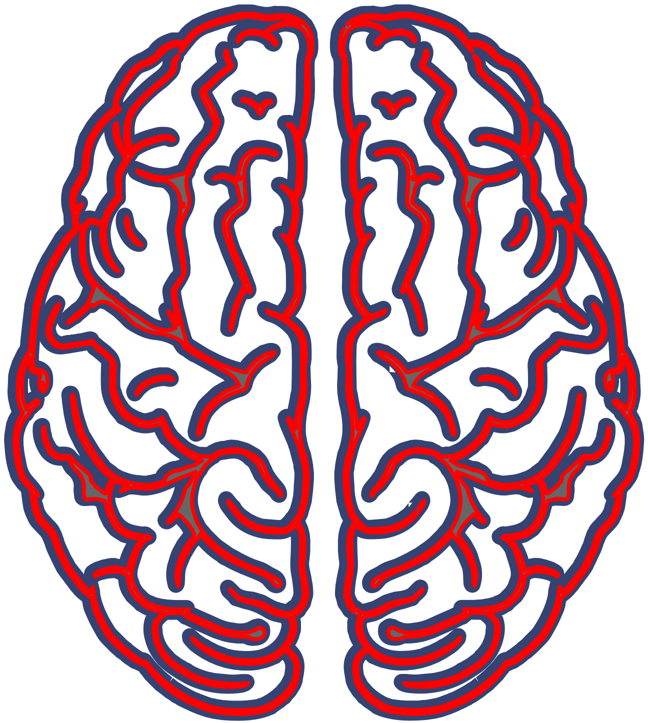
EPFL



MINI PROJECT 1:

Comparative fMRI Analysis of Hand and Foot Movement Representations

Angana Mondal, Arnault Stähli, Heliya Shakeri, Khushi Singh, Pamela van den Enden

NX421:
NEURAL SIGNALS AND SIGNAL PROCESSING

Prof. Van de Ville

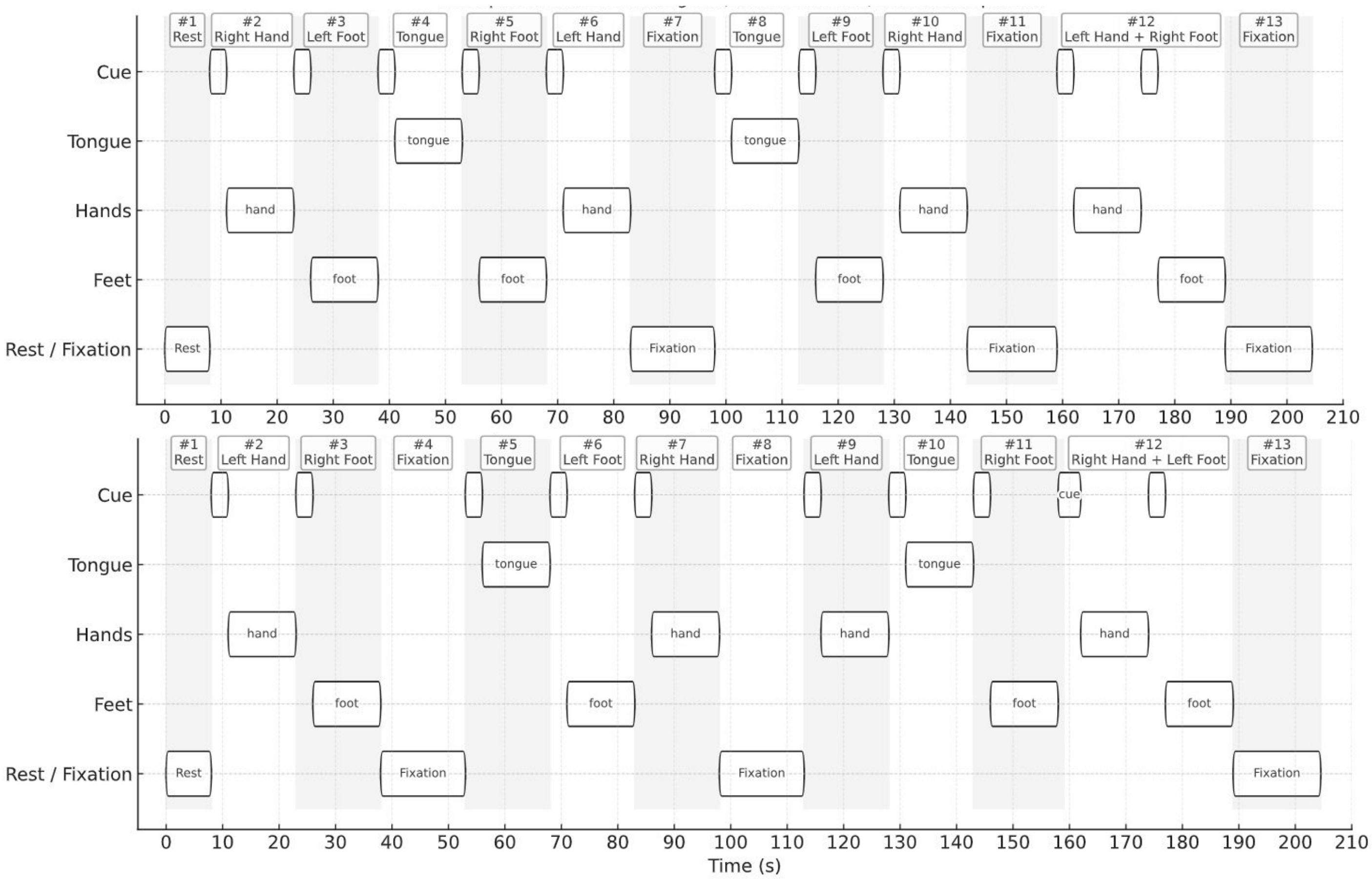
MOTIVATION

To compare the neural activation during motion of **hands vs feet** on a visual cue, using BOLD response in **fMRI** data.

DATASET

Run 1

Run 2



STRUCTURAL PREPROCESSING

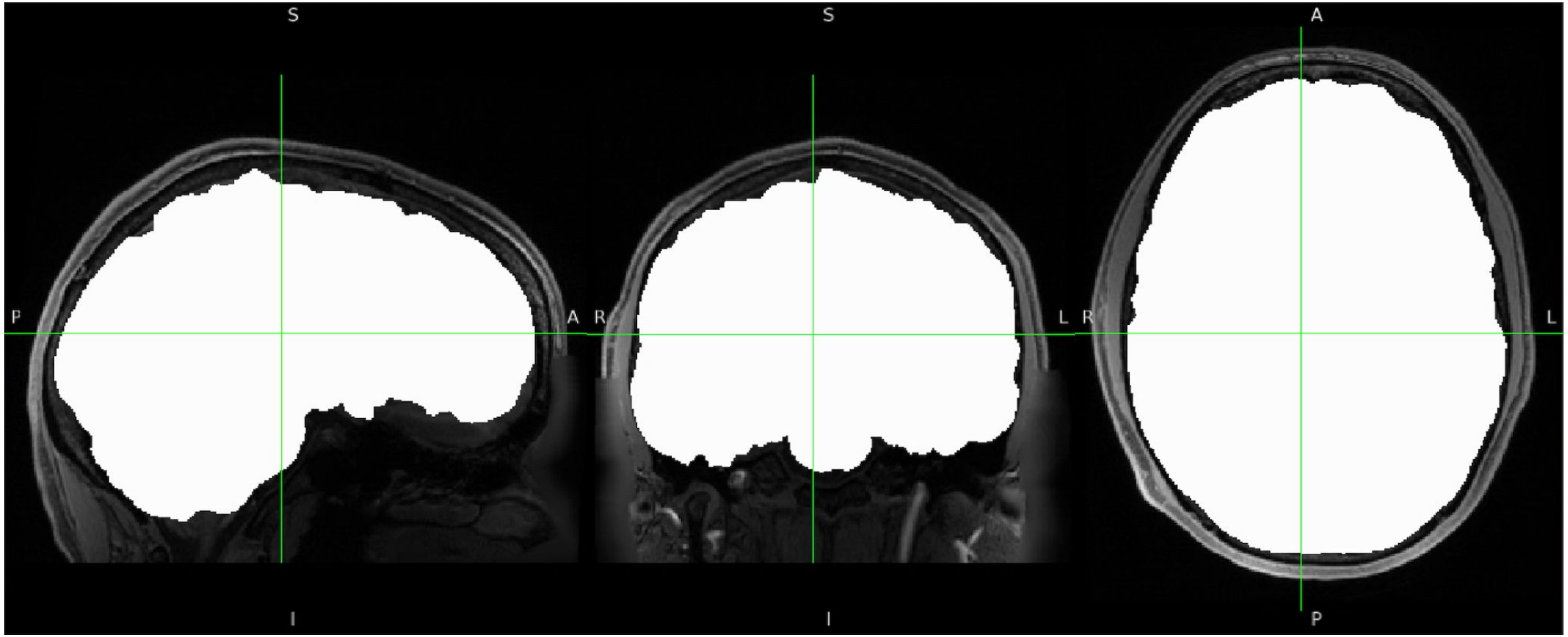
Skull Stripping

Segmentation

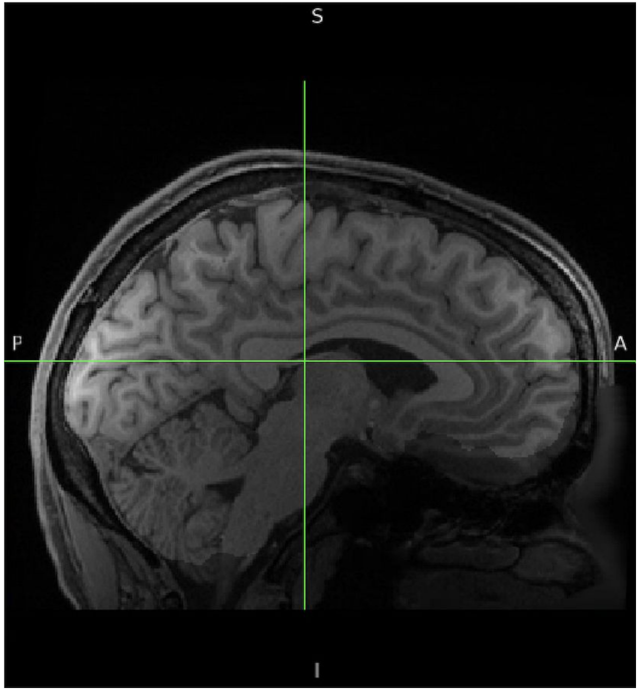
SKULL STRIPPING

Removed non-brain tissues from the anatomical image to isolate the brain for further processing.

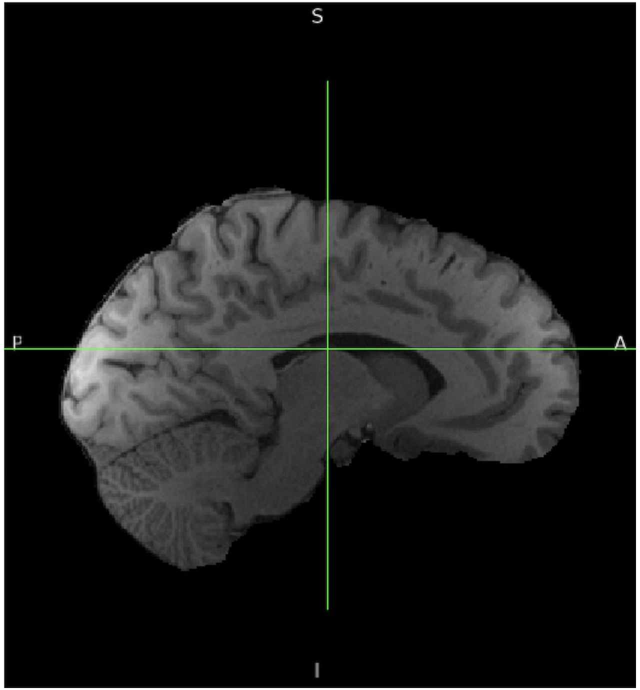
MASK ALIGNMENT CONFIGURATION



BEFORE SKULL STRIPPING



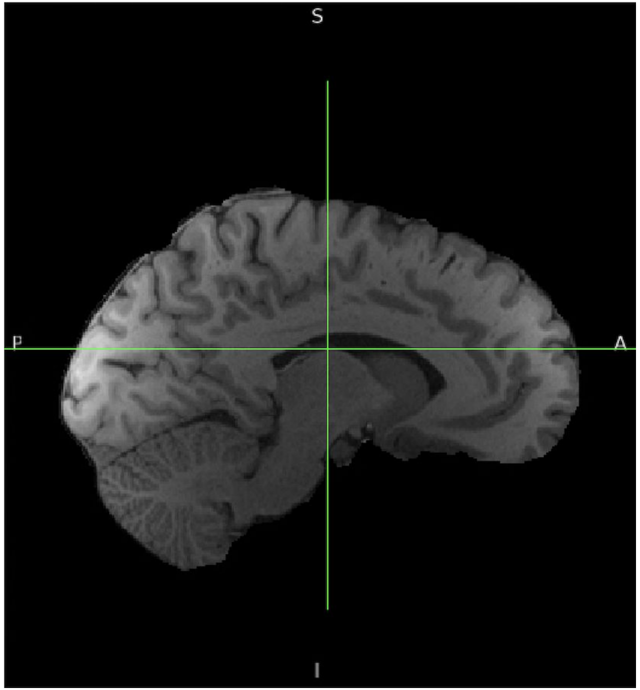
AFTER SKULL STRIPPING



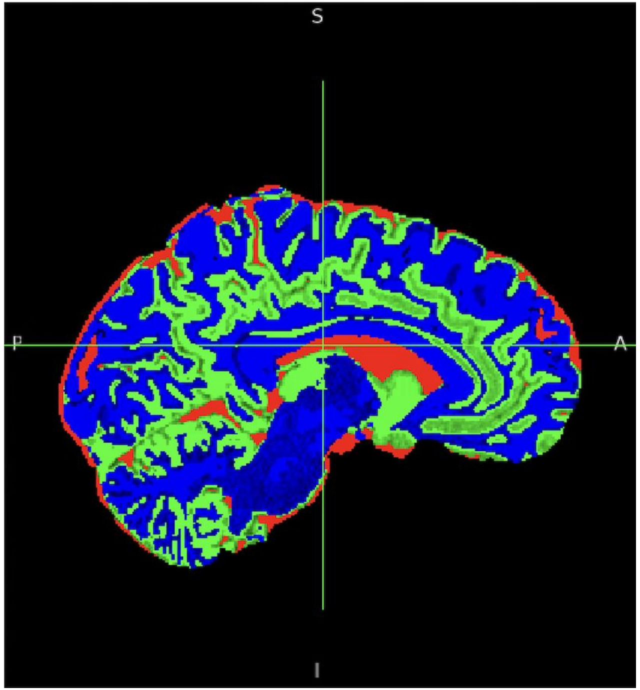
SEGMENTATION

Separated the brain into tissue types:
gray matter, white matter, CSF

BEFORE SEGMENTATION



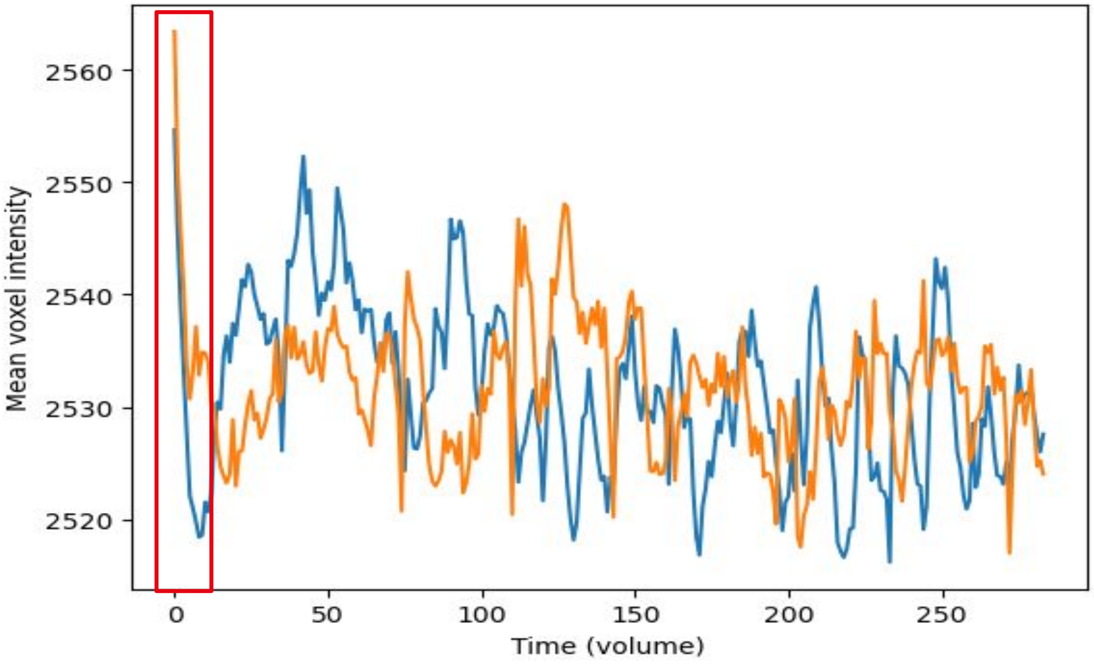
AFTER SEGMENTATION



FUNCTIONAL PREPROCESSING

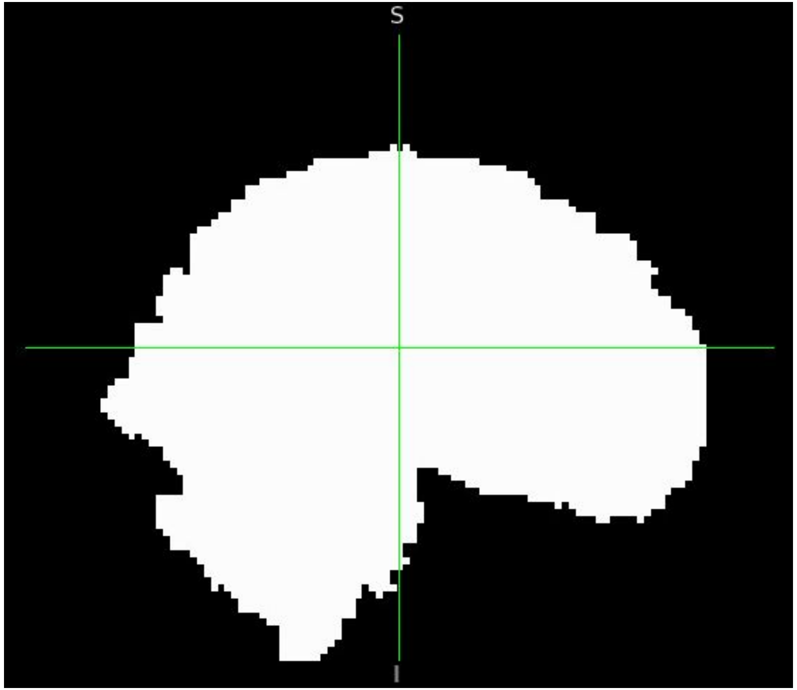
Checking for Problematic Volumes
Variance Normalisation
Motion Correction
Coregistration
Gaussian Smoothing

CHECKING FOR PROBLEMATIC VOLUMES

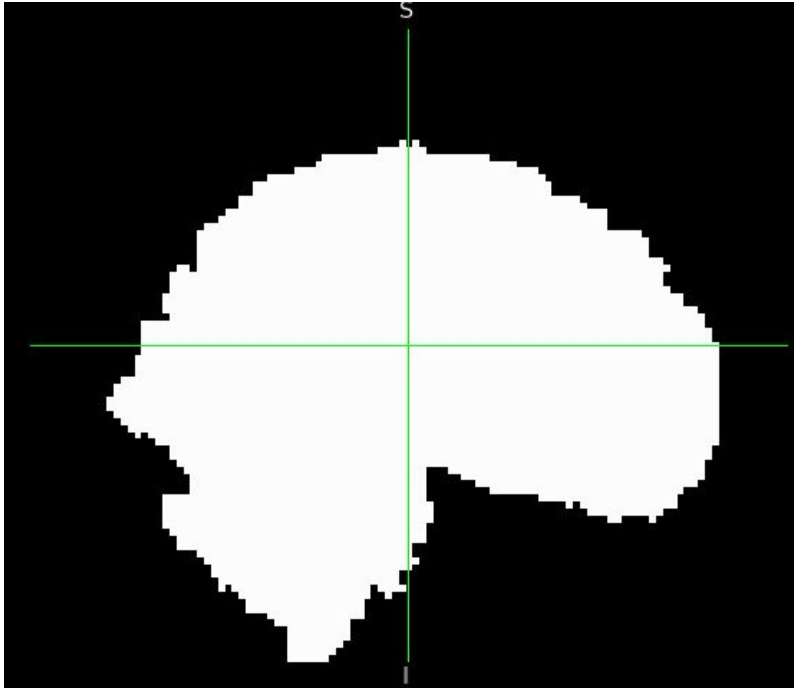


MASKING

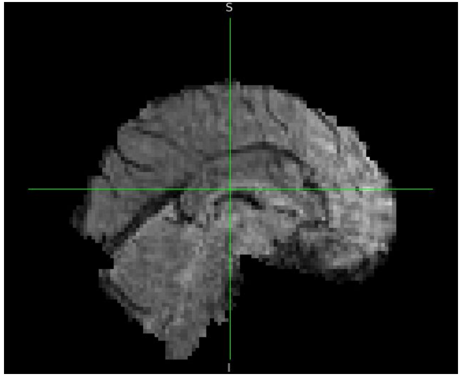
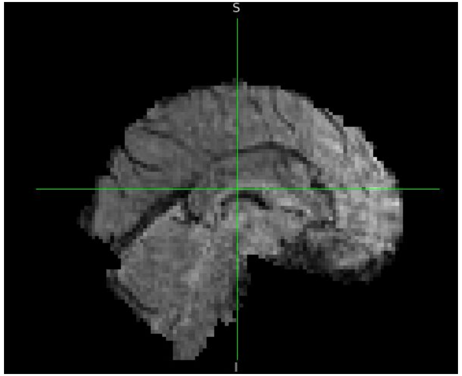
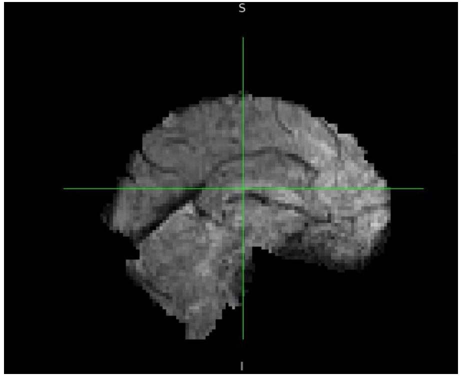
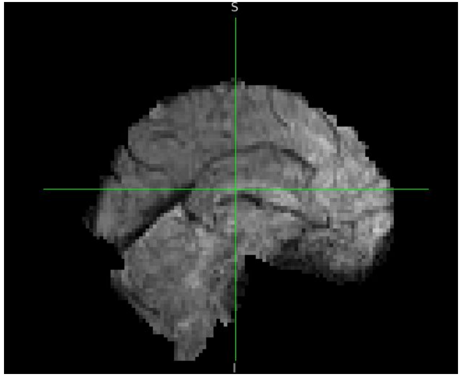
MASK (LR)



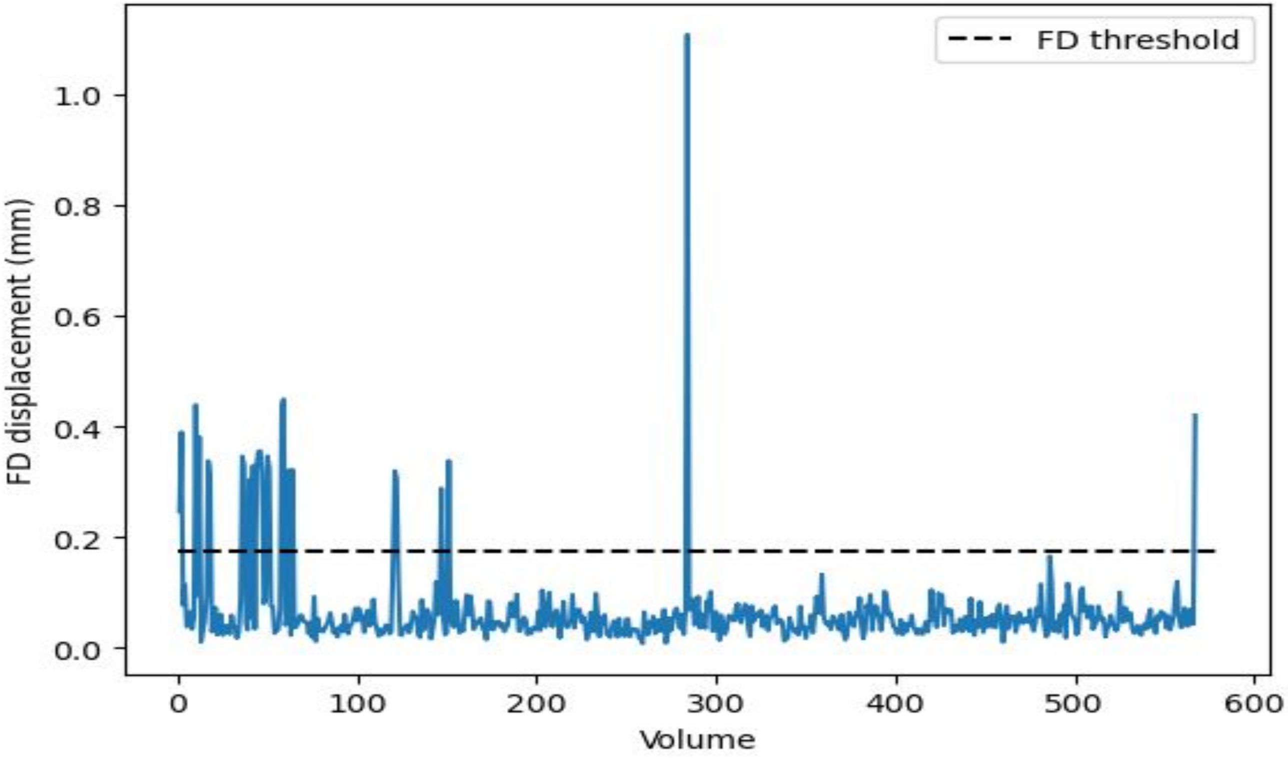
MASK (RL)



VARIANCE NORMALIZATION

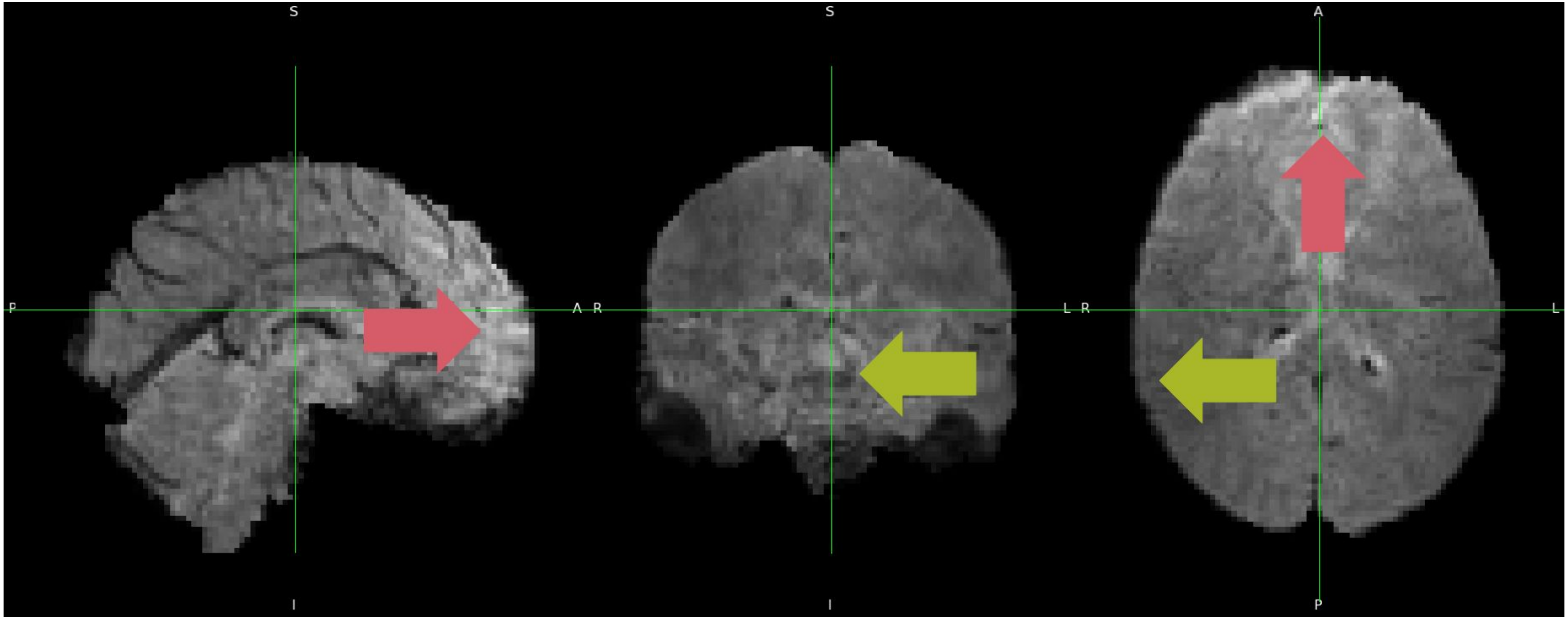
	BEFORE NORMALISATION	AFTER NORMALISATION
RUN 1 (LR)		
RUN 2 (RL)		

MOTION CORRECTION

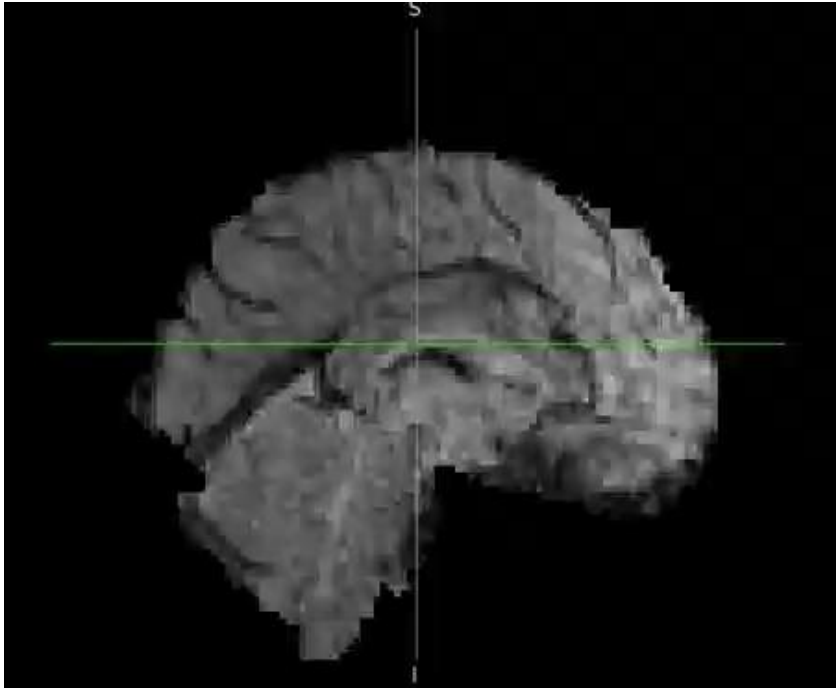


$$\text{Threshold} = 2 \times Q_{0.75} + 1.5 \times (Q_{0.75} - Q_{0.25})$$

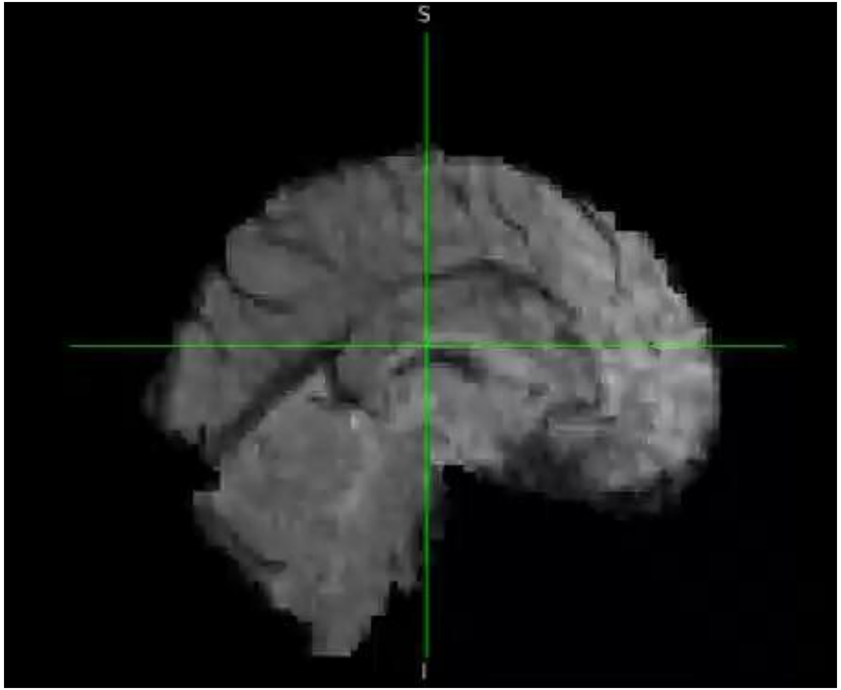
EFFECTS OF MOTION CORRECTION



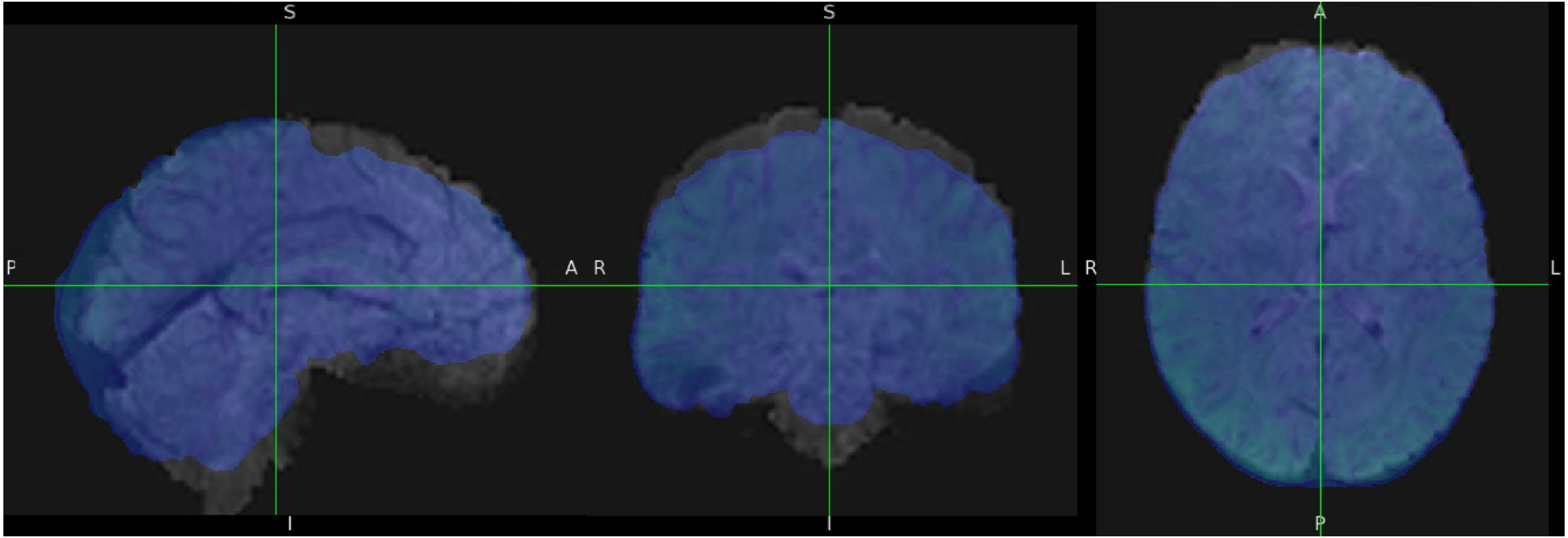
BEFORE MOTION CORRECTION



AFTER MOTION CORRECTION



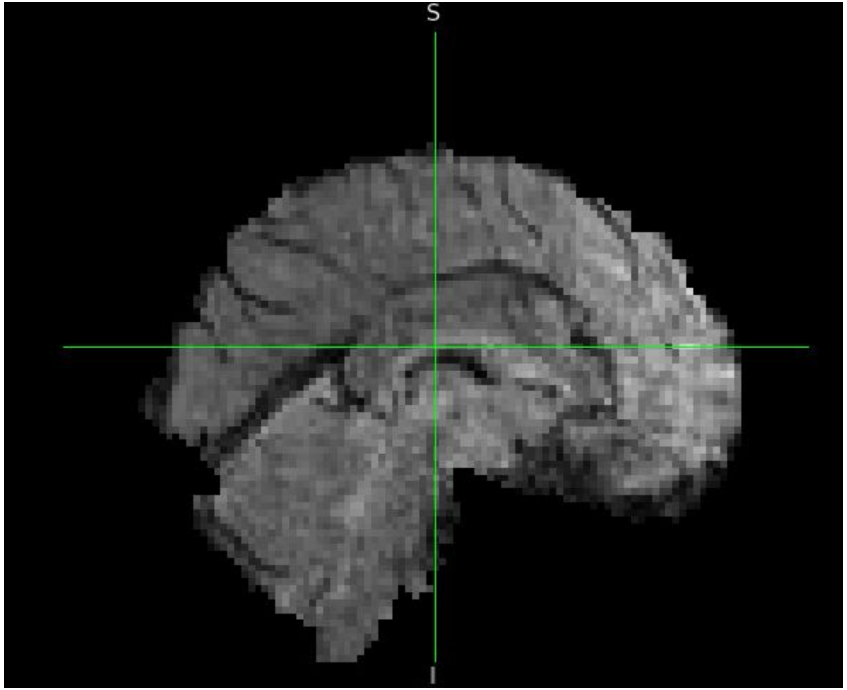
COREGISTRATION



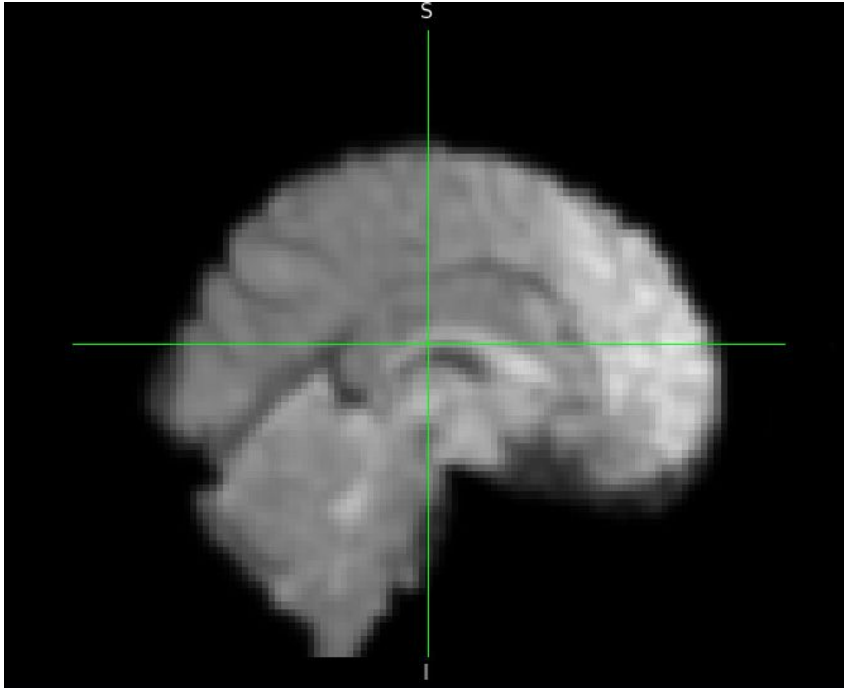
❑ EPI Space - FMRI volume
❑ T1 Space - Structural MRI volume

GAUSSIAN SMOOTHING

BEFORE GAUSSIAN SMOOTHING



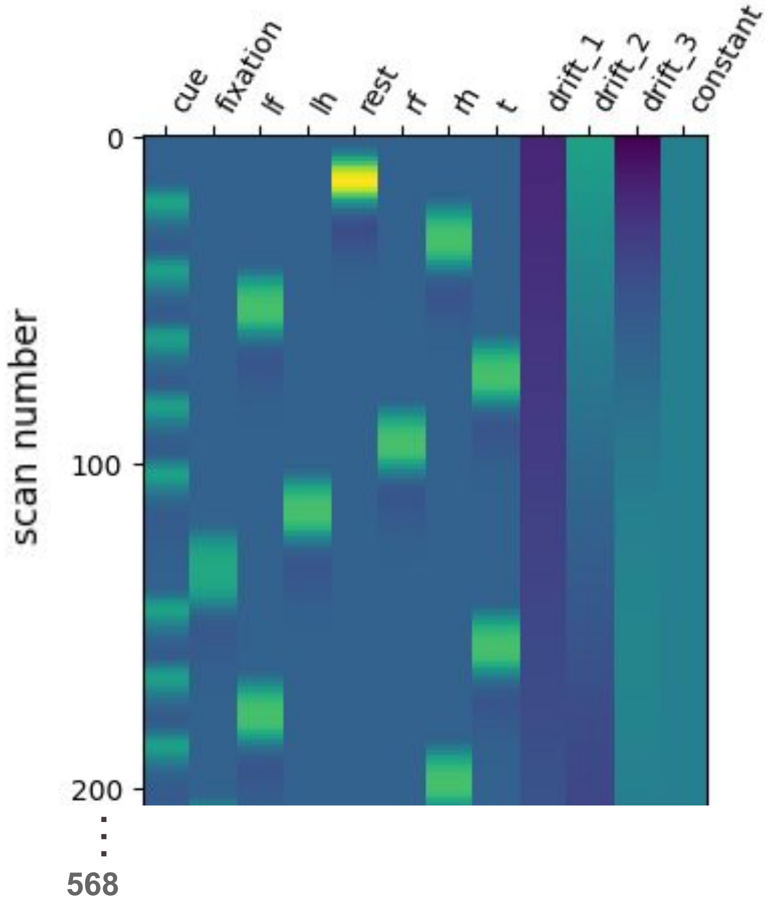
AFTER GAUSSIAN SMOOTHING



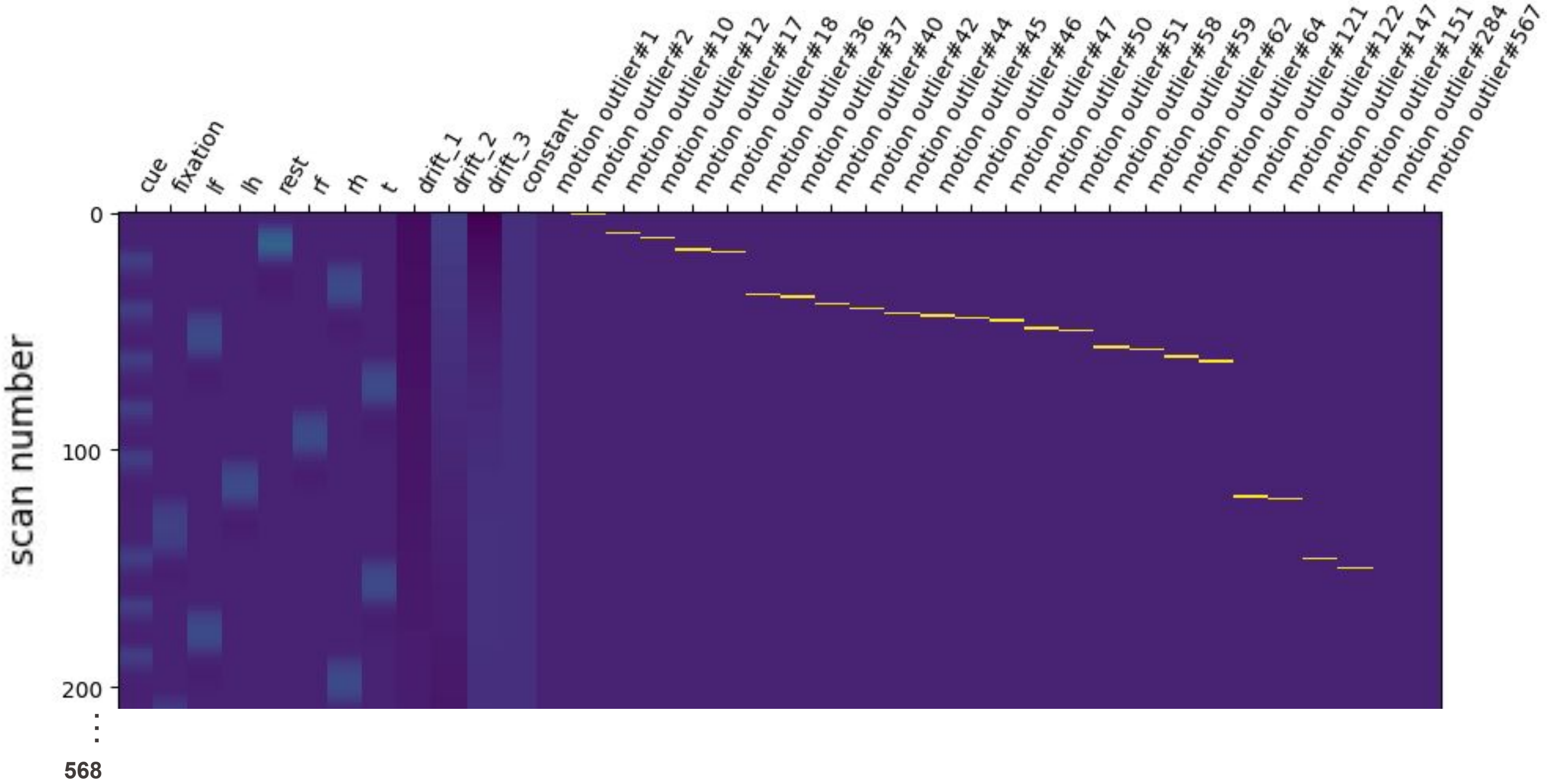
GENERAL LINEAR MODEL (GLM)

Design Matrix
Statistical Maps
Activation Maps
AAL Atlas Overlay

DESIGN MATRIX



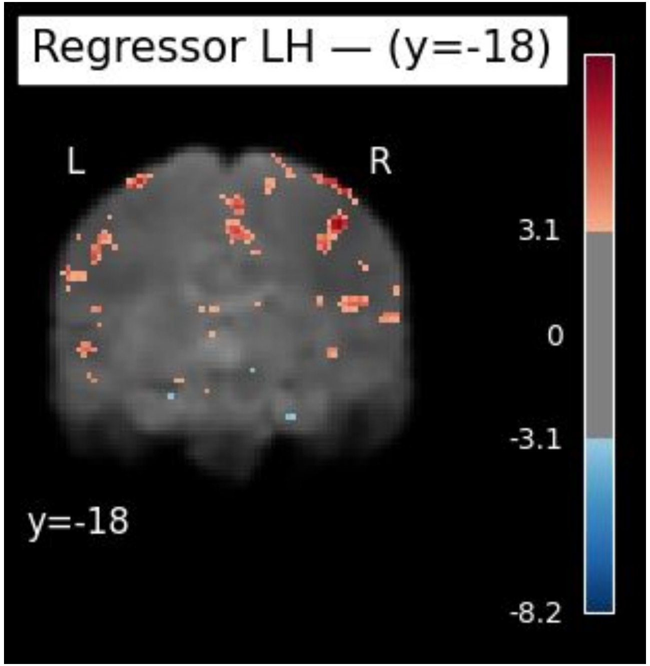
ADJUSTED DESIGN MATRIX



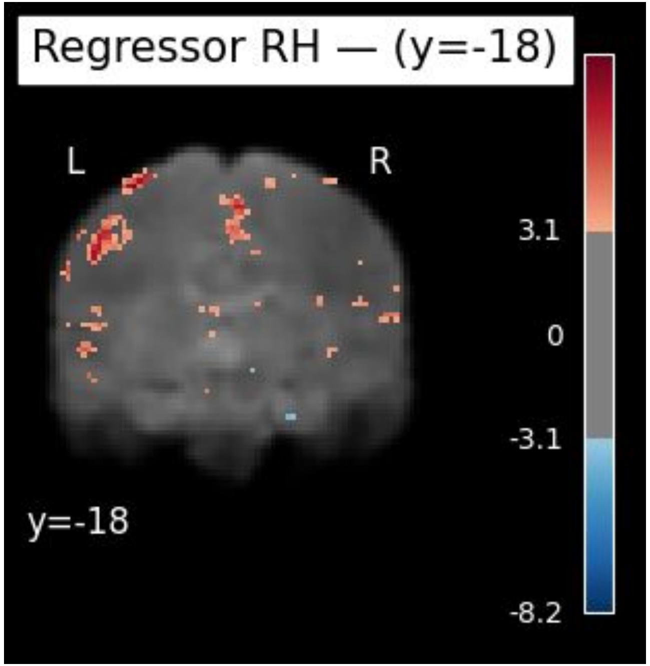
STATISTICAL MAPS

HANDS

LEFT



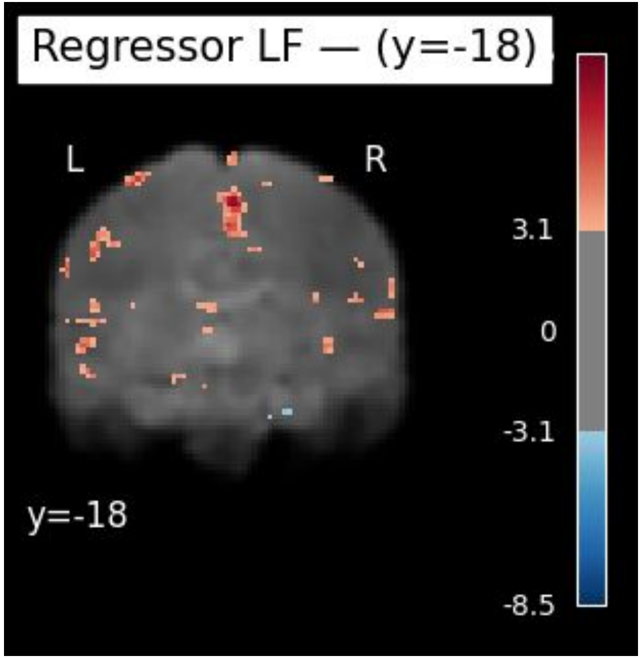
RIGHT



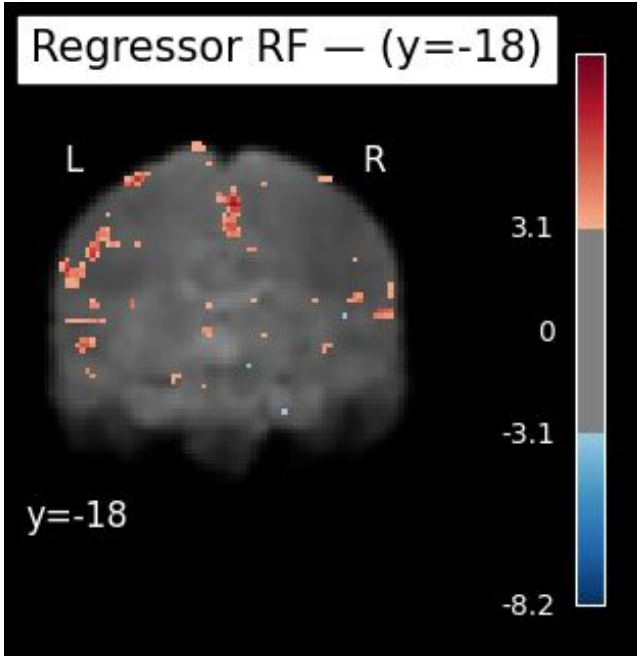
STATISTICAL MAPS

FEET

LEFT

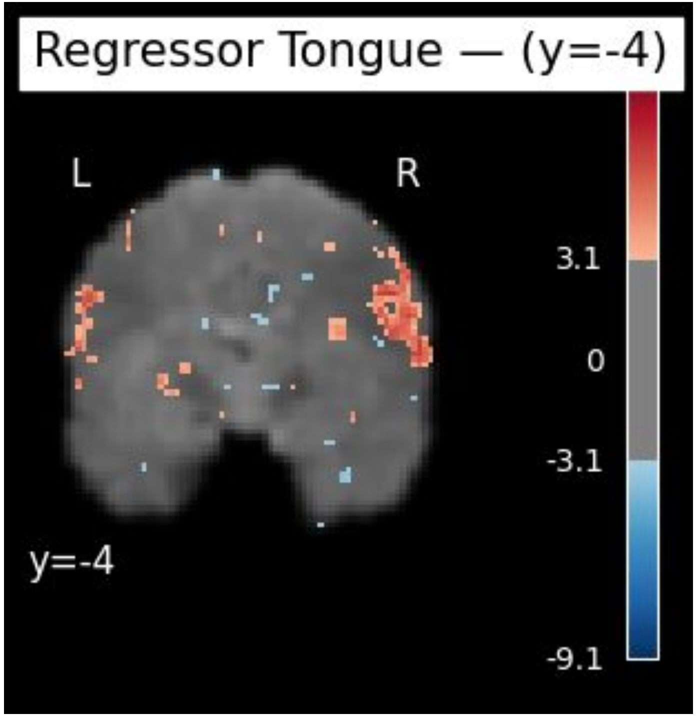


RIGHT

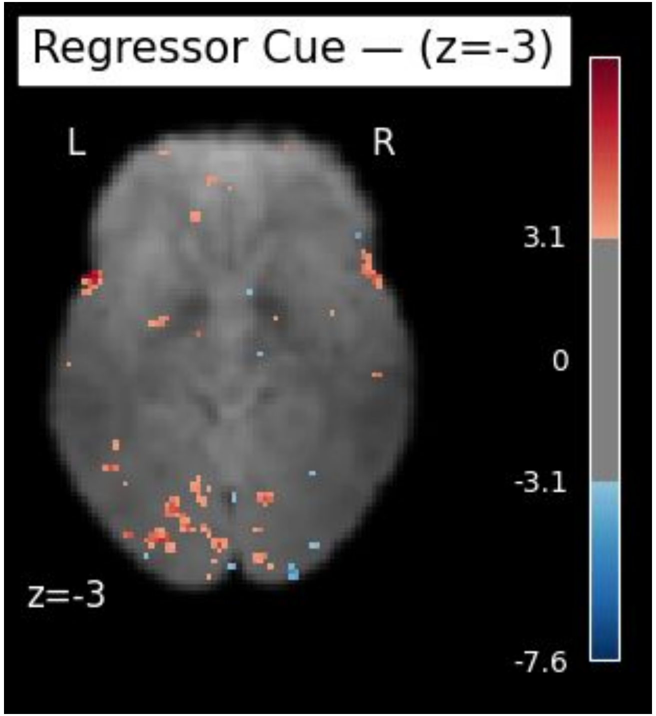


STATISTICAL MAPS

TONGUE



CUE



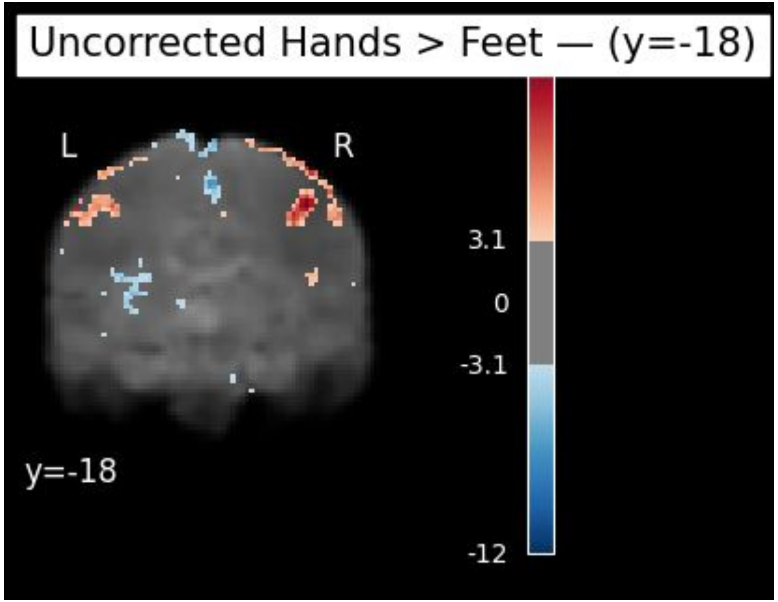
CONTRAST VECTOR

Average Hands > Average Feet

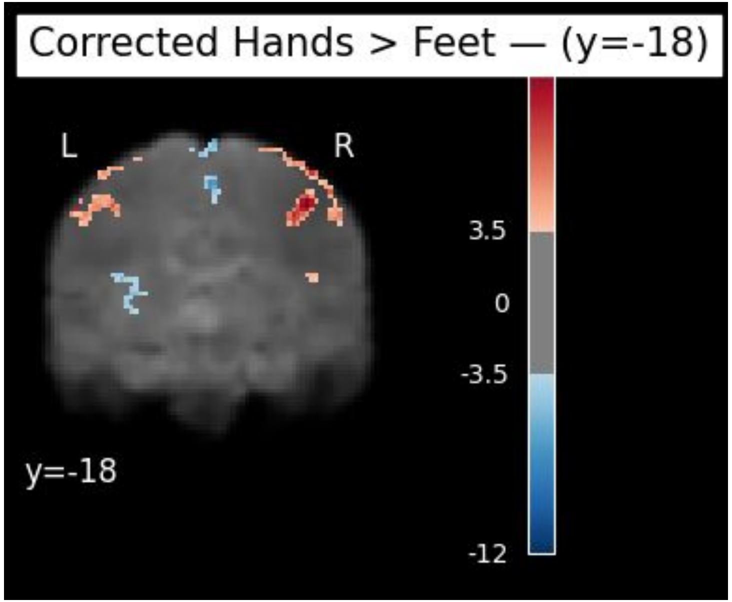
$\frac{1}{2}$(Right Hand + Left Hand) - $\frac{1}{2}$ (Right Foot + Left Foot)

FDR AND CLUSTER CORRECTION

BEFORE CORRECTION

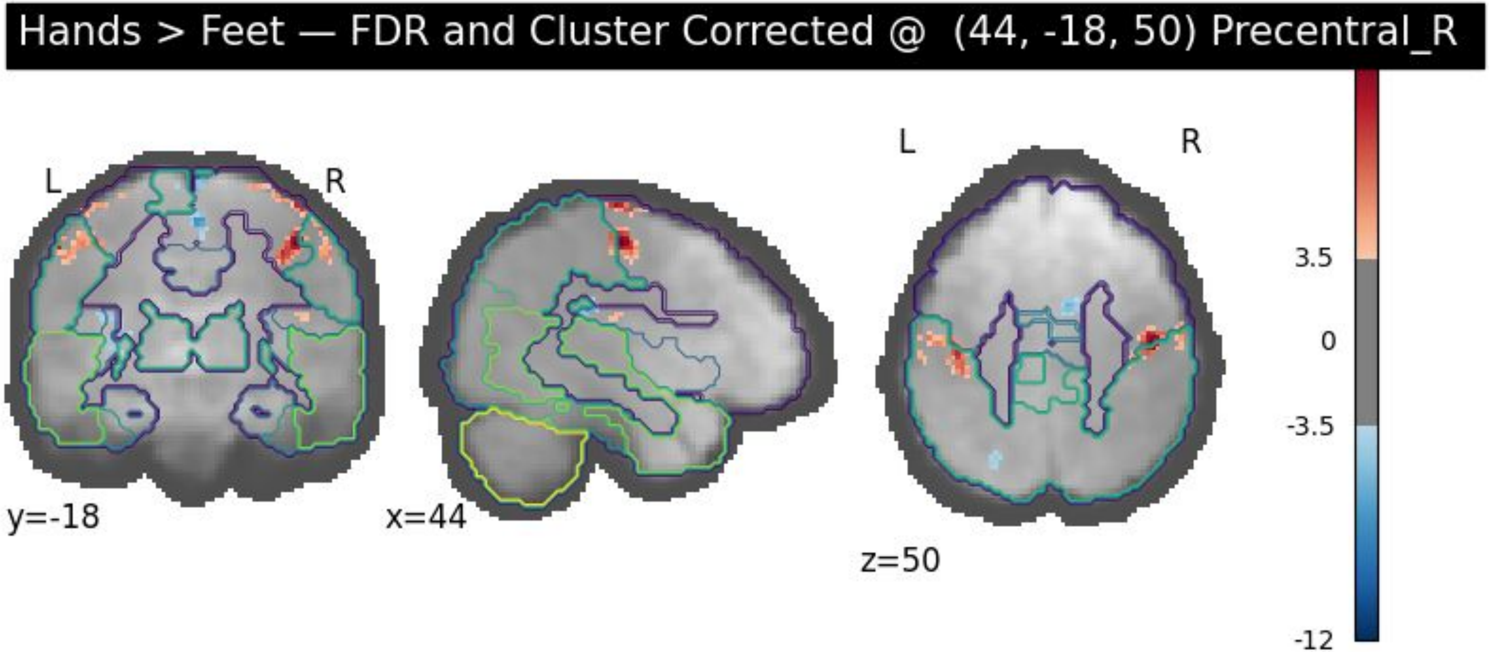


AFTER CORRECTION

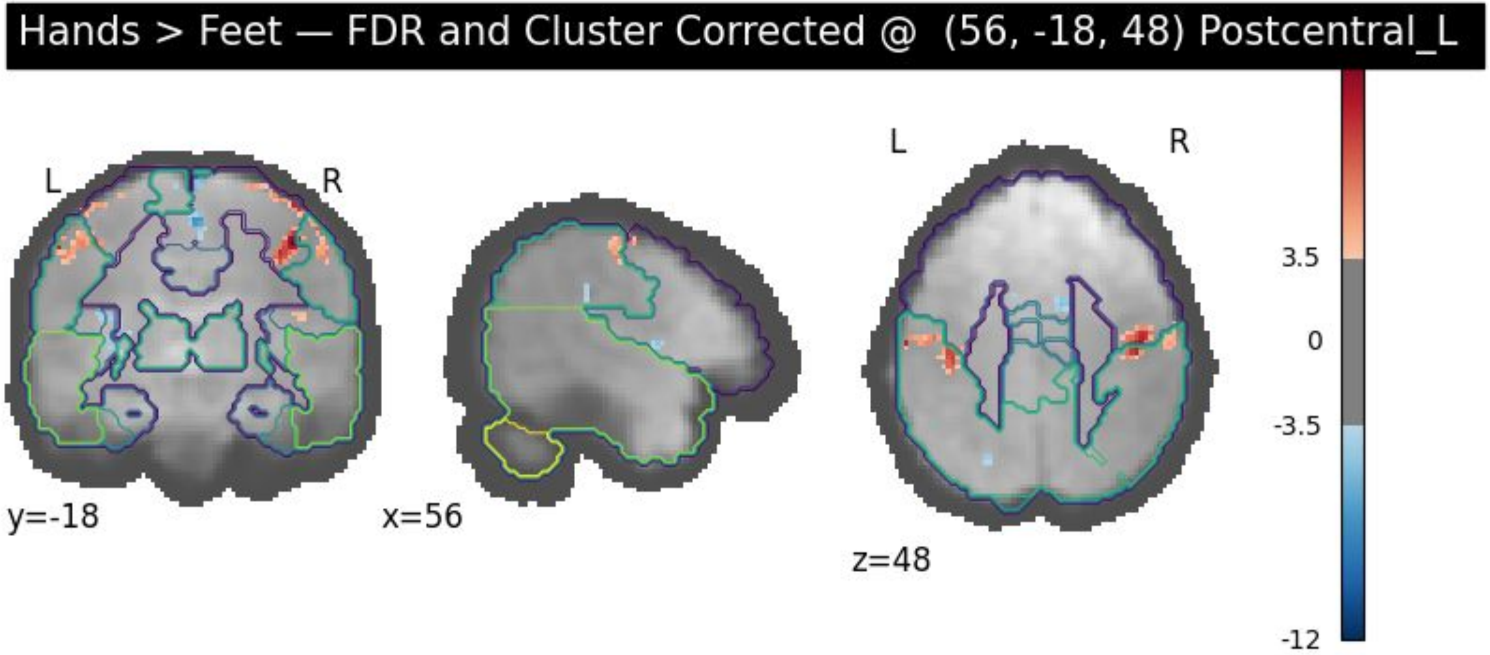


$q < 0.05$, cluster > 10 voxels

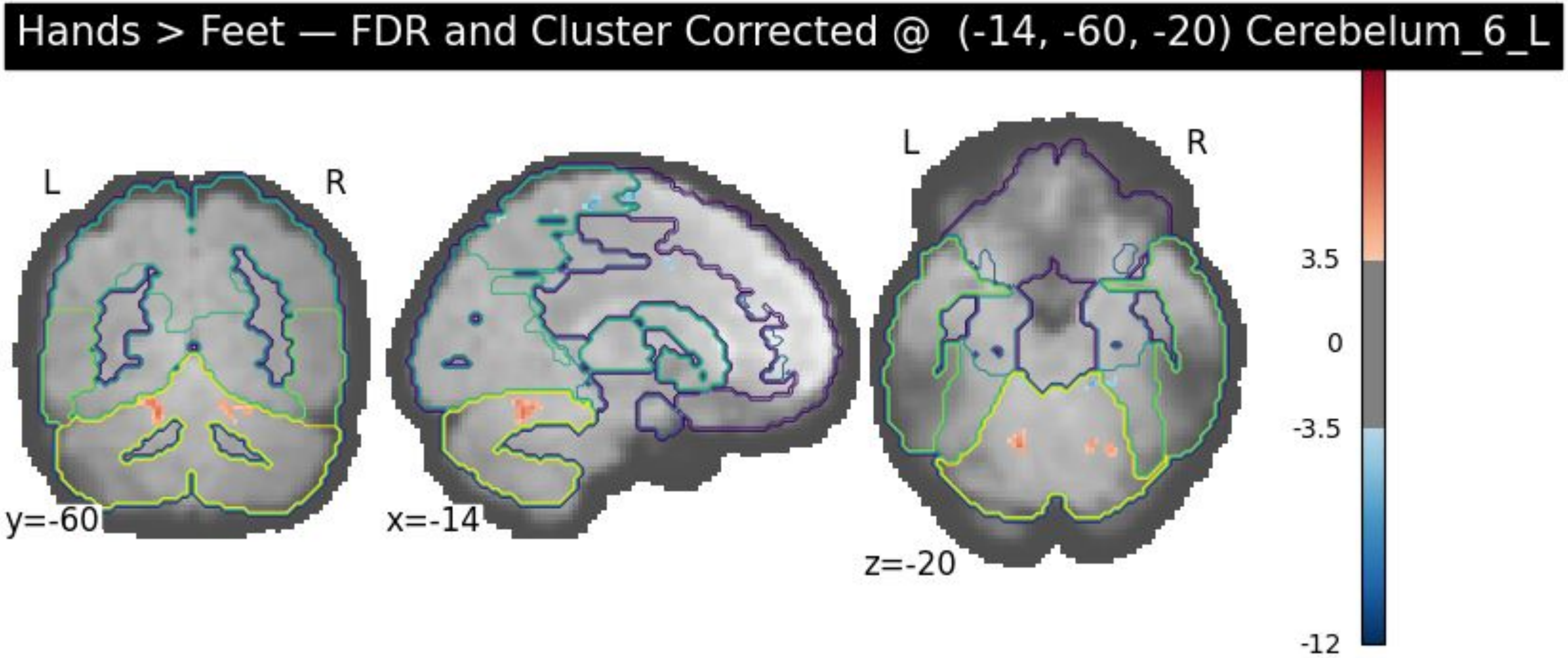
HAND > FEET: AAL ATLAS OVERLAY
(z = 12.30)



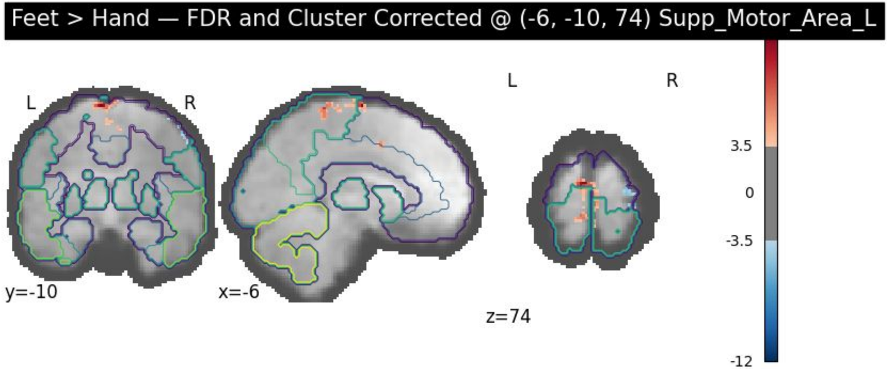
HAND > FEET: AAL ATLAS OVERLAY
(z = 9.57)

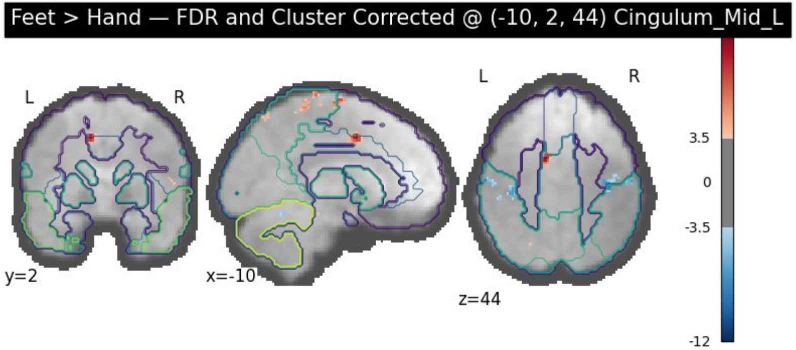


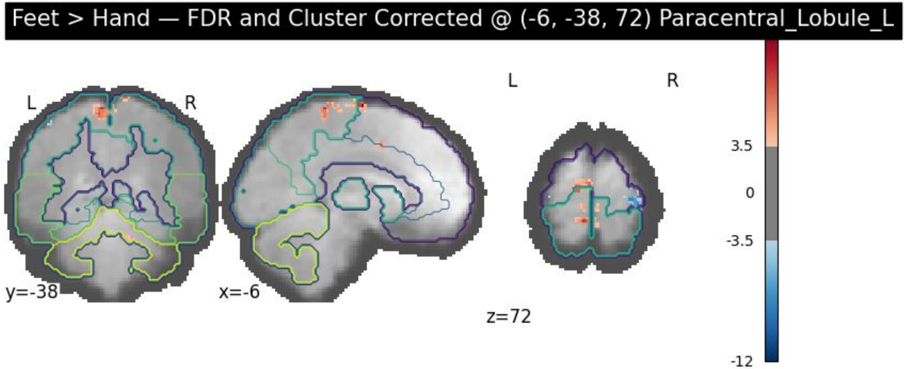
HAND > FEET: AAL ATLAS OVERLAY
(z = 6.65)

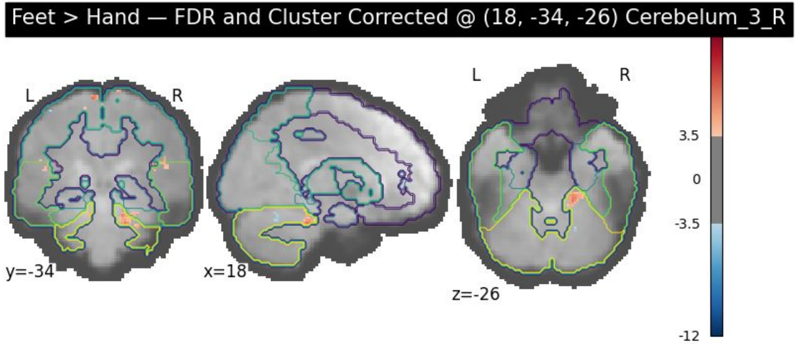


FEET > HANDS: ATLAS OVERLAY







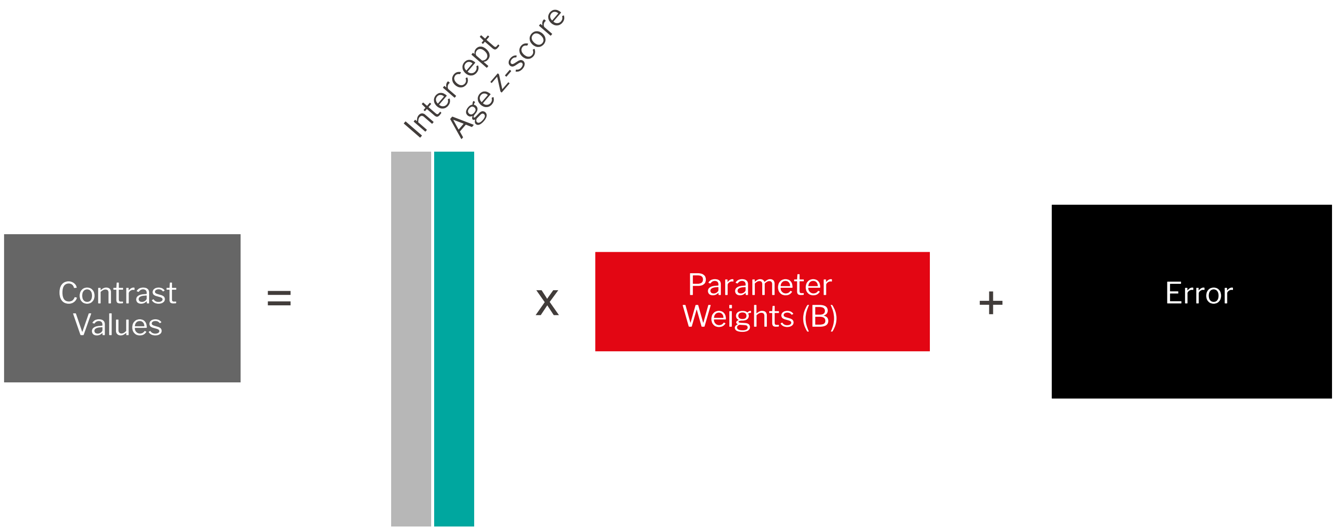


THEORETICAL QUESTIONS PART 1

Scope of **Second Level Analysis** on:

1) provided dataset
2) extended 10 subject dataset + subject ages

SECOND-LEVEL GLM

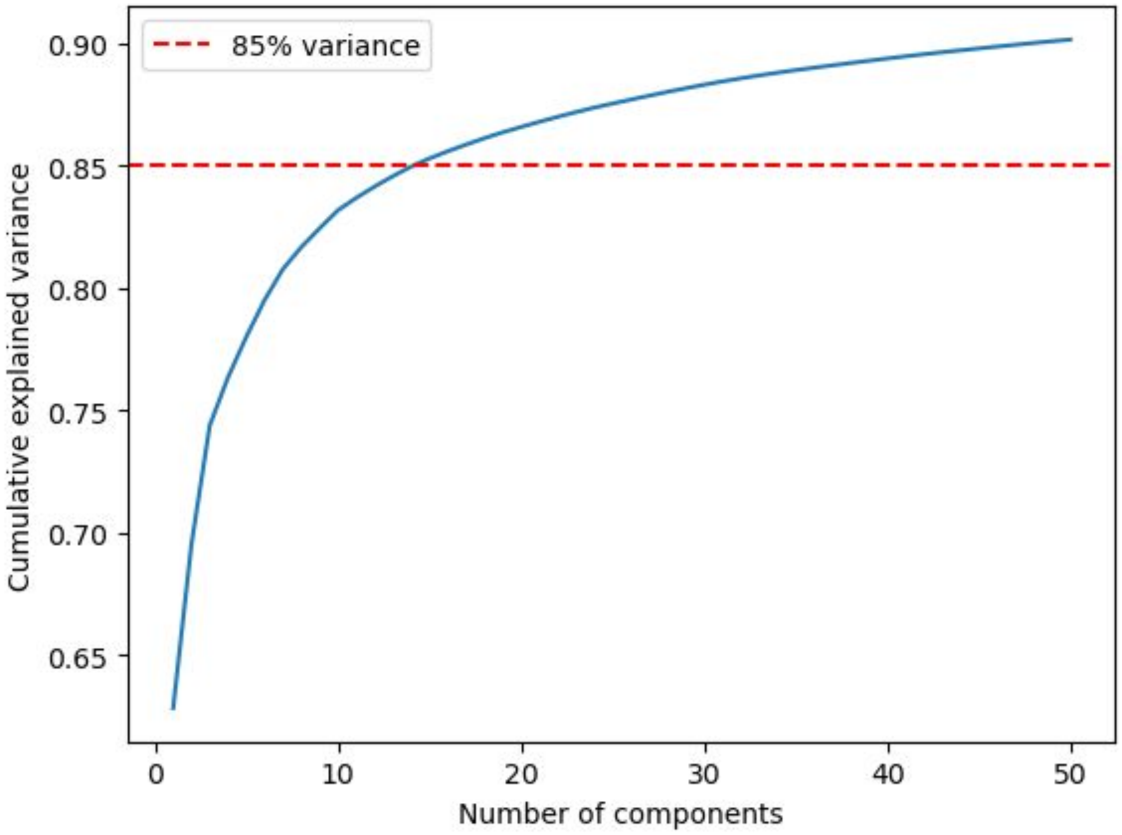


Does hand-vs-feet activation vary with age? **CONTRAST: [0 1]**

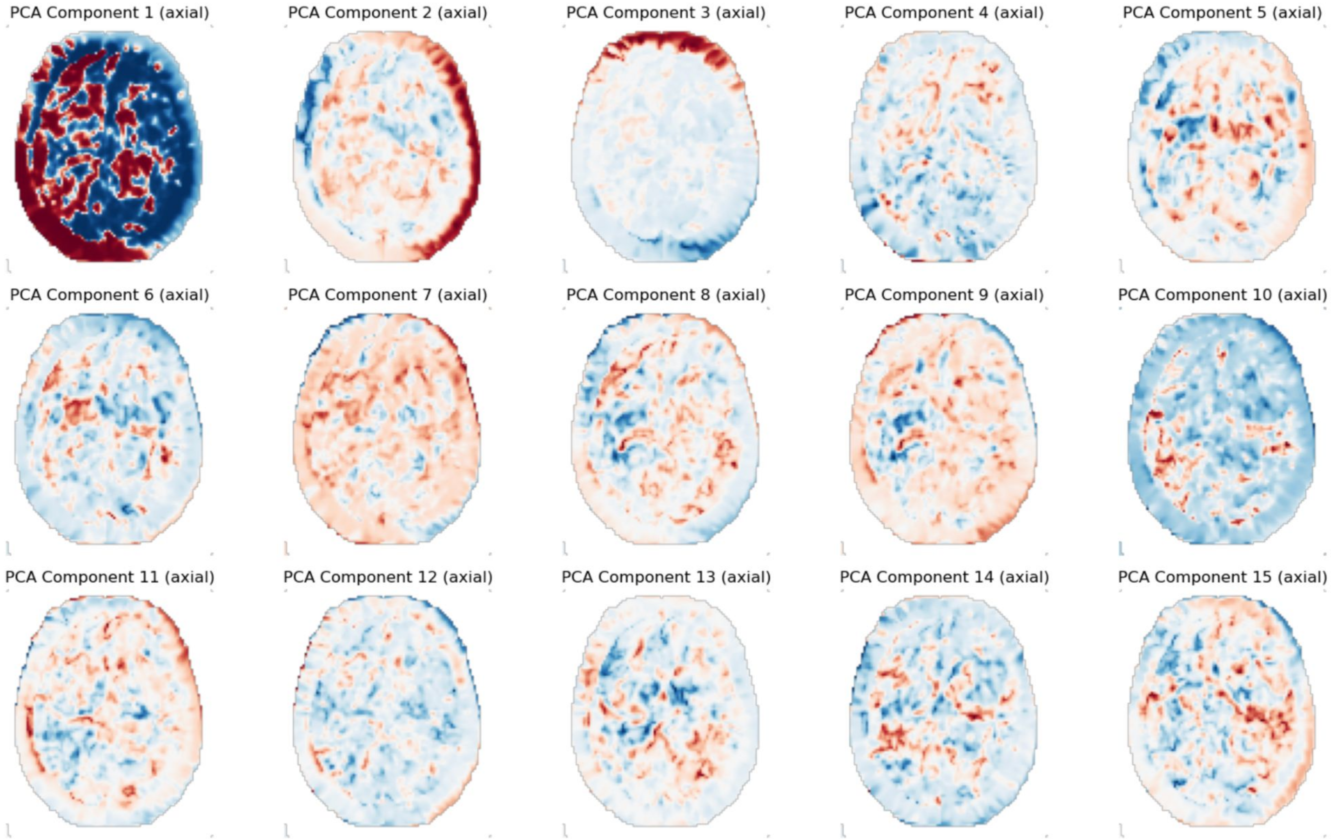
PRINCIPAL COMPONENT ANALYSIS (PCA)

Decomposition
Component Selection
Visualisation
Similarity Matrix

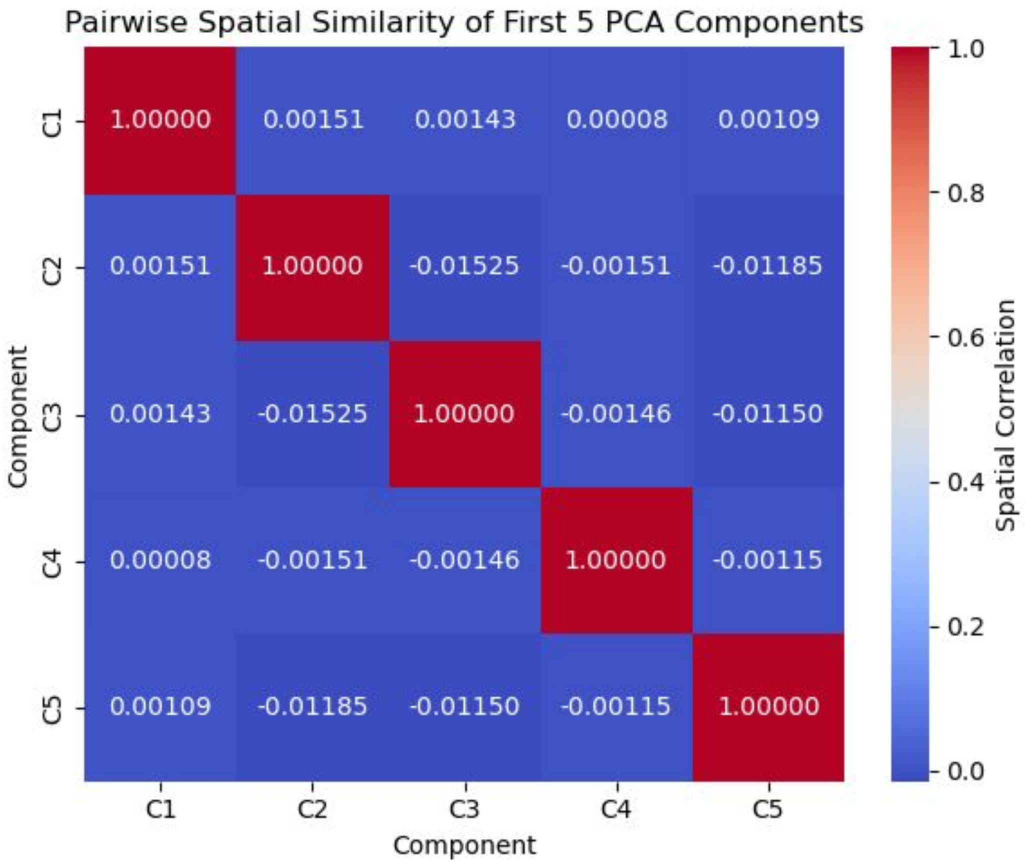
COMPONENT SELECTION



PCA COMPONENTS OVERLAID ON ANATOMY



SIMILARITY MATRIX

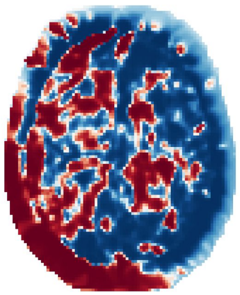


THEORETICAL QUESTIONS PART 2

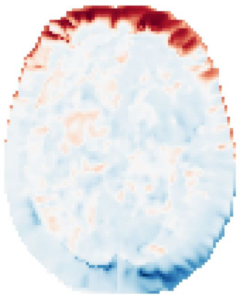
Identify Functional Networks
Comparison with GLM: results, applications

FUNCTIONAL NETWORKS

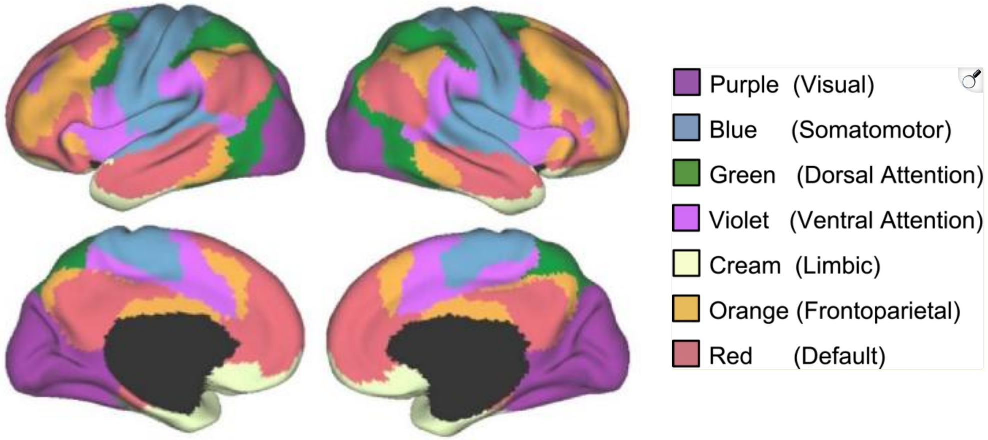
PC 1: Contralateral Control



PC 3: Visual Network



COARSE PARCELLATION
7 FUNCTIONAL NETWORKS

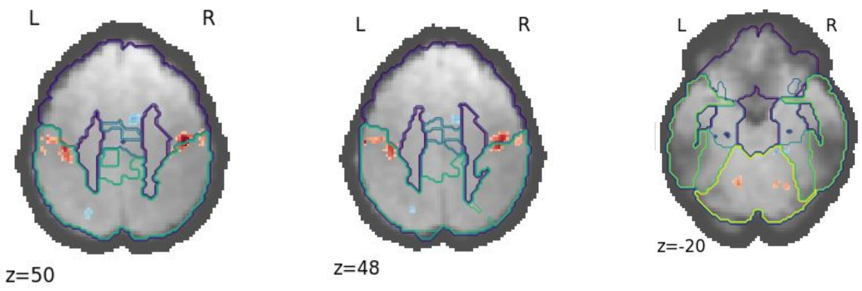


Yeo et al., 2011

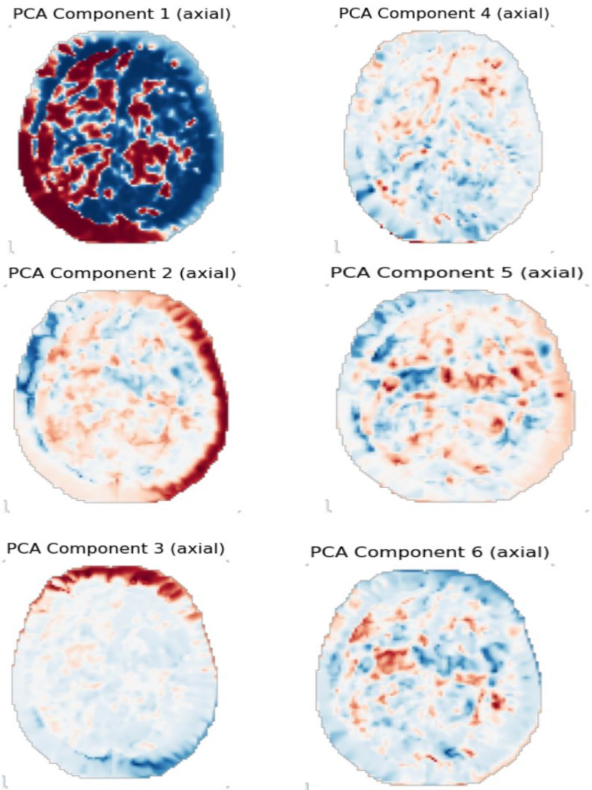
GLM vs PCA

GLM results

Precentral Gyrus (M1) Postcentral Gyrus (S1) Cerebellum



PCA results



TEAM CONTRIBUTIONS

ANGANA MONDAL - Coregistration, PCA
ARNAULT STÄHLI - Functional Preprocessing
HELIYA SHAKERI - Structural Preprocessing
KHUSHI SINGH - Gaussian Smoothing, PCA, slides
PAMELA VAN DEN ENDEN - General Linear Model

THANK YOU